



GRN10

Variável A

$$\frac{dA}{dt} = \frac{1}{\tau_A} \left(1 - \frac{N(J)^{n_{AJ}}}{N(J)^{n_{AJ}} + k_{AJ}^{n_{AJ}}} - N(A) \right)$$

Variável B

$$\frac{dB}{dt} = \frac{1}{\tau_B} \left(\frac{N(E)^{n_{BE}}}{N(E)^{n_{BE}} + k_{BE}^{n_{BE}}} - N(B) \right)$$

Variável C

$$\frac{dC}{dt} = \frac{1}{\tau_C} \left(\begin{aligned} & \left(\frac{N(B)^{n_{CB}}}{N(B)^{n_{CB}} + k_{CB}^{n_{CB}}} \cdot \left(1 - \frac{N(F)^{n_{CF}}}{N(F)^{n_{CF}} + k_{CF}^{n_{CF}}} \right) \cdot \left(1 - \frac{N(A)^{n_{CA}}}{N(A)^{n_{CA}} + k_{CA}^{n_{CA}}} \right) \right) + \\ & \left(\left(1 - \frac{N(B)^{n_{CB}}}{N(B)^{n_{CB}} + k_{CB}^{n_{CB}}} \right) \cdot \frac{N(F)^{n_{CF}}}{N(F)^{n_{CF}} + k_{CF}^{n_{CF}}} \cdot \left(1 - \frac{N(A)^{n_{CA}}}{N(A)^{n_{CA}} + k_{CA}^{n_{CA}}} \right) \right) + \\ & \left(\left(1 - \frac{N(B)^{n_{CB}}}{N(B)^{n_{CB}} + k_{CB}^{n_{CB}}} \right) \cdot \left(1 - \frac{N(F)^{n_{CF}}}{N(F)^{n_{CF}} + k_{CF}^{n_{CF}}} \right) \cdot \frac{N(A)^{n_{CA}}}{N(A)^{n_{CA}} + k_{CA}^{n_{CA}}} \right) + \\ & \left(\frac{N(B)^{n_{CB}}}{N(B)^{n_{CB}} + k_{CB}^{n_{CB}}} \cdot \left(1 - \frac{N(F)^{n_{CF}}}{N(F)^{n_{CF}} + k_{CF}^{n_{CF}}} \right) \cdot \frac{N(A)^{n_{CA}}}{N(A)^{n_{CA}} + k_{CA}^{n_{CA}}} \right) + \\ & \left(\left(1 - \frac{N(B)^{n_{CB}}}{N(B)^{n_{CB}} + k_{CB}^{n_{CB}}} \right) \cdot \frac{N(F)^{n_{CF}}}{N(F)^{n_{CF}} + k_{CF}^{n_{CF}}} \cdot \frac{N(A)^{n_{CA}}}{N(A)^{n_{CA}} + k_{CA}^{n_{CA}}} \right) + \\ & \left(\frac{N(B)^{n_{CB}}}{N(B)^{n_{CB}} + k_{CB}^{n_{CB}}} \cdot \frac{N(F)^{n_{CF}}}{N(F)^{n_{CF}} + k_{CF}^{n_{CF}}} \cdot \frac{N(A)^{n_{CA}}}{N(A)^{n_{CA}} + k_{CA}^{n_{CA}}} \right) \end{aligned} \right) - N(C)$$

Variável D

$$\frac{dD}{dt} = \frac{1}{\tau_D} \left(\frac{N(F)^{n_{DF}}}{N(F)^{n_{DF}} + k_{DF}^{n_{DF}}} - N(D) \right)$$

Variável E

$$\frac{dE}{dt} = \frac{1}{\tau_E} \left(1 - \frac{N(J)^{n_{EJ}}}{N(J)^{n_{EJ}} + k_{EJ}^{n_{EJ}}} - N(E) \right)$$

Variável F

$$\frac{dF}{dt} = \frac{1}{\tau_F} \left(\frac{N(A)^{n_{FA}}}{N(A)^{n_{FA}} + k_{FA}^{n_{FA}}} - N(F) \right)$$

Variável G

$$\frac{dG}{dt} = \frac{1}{\tau_G} \left(\begin{aligned} & \left(\frac{N(B)^{n_{GB}}}{N(B)^{n_{GB}} + k_{GB}^{n_{GB}}} \cdot \left(1 - \frac{N(F)^{n_{GF}}}{N(F)^{n_{GF}} + k_{GF}^{n_{GF}}} \right) \cdot \left(1 - \frac{N(A)^{n_{GA}}}{N(A)^{n_{GA}} + k_{GA}^{n_{GA}}} \right) \right) + \\ & \left(\left(1 - \frac{N(B)^{n_{GB}}}{N(B)^{n_{GB}} + k_{GB}^{n_{GB}}} \right) \cdot \frac{N(F)^{n_{GF}}}{N(F)^{n_{GF}} + k_{GF}^{n_{GF}}} \cdot \left(1 - \frac{N(A)^{n_{GA}}}{N(A)^{n_{GA}} + k_{GA}^{n_{GA}}} \right) \right) + \\ & \left(\left(1 - \frac{N(B)^{n_{GB}}}{N(B)^{n_{GB}} + k_{GB}^{n_{GB}}} \right) \cdot \left(1 - \frac{N(F)^{n_{GF}}}{N(F)^{n_{GF}} + k_{GF}^{n_{GF}}} \right) \cdot \frac{N(A)^{n_{GA}}}{N(A)^{n_{GA}} + k_{GA}^{n_{GA}}} \right) + \\ & \left(\frac{N(B)^{n_{GB}}}{N(B)^{n_{GB}} + k_{GB}^{n_{GB}}} \cdot \left(1 - \frac{N(F)^{n_{GF}}}{N(F)^{n_{GF}} + k_{GF}^{n_{GF}}} \right) \cdot \frac{N(A)^{n_{GA}}}{N(A)^{n_{GA}} + k_{GA}^{n_{GA}}} \right) + \\ & \left(\left(1 - \frac{N(B)^{n_{GB}}}{N(B)^{n_{GB}} + k_{GB}^{n_{GB}}} \right) \cdot \frac{N(F)^{n_{GF}}}{N(F)^{n_{GF}} + k_{GF}^{n_{GF}}} \cdot \frac{N(A)^{n_{GA}}}{N(A)^{n_{GA}} + k_{GA}^{n_{GA}}} \right) + \\ & \left(\frac{N(B)^{n_{GB}}}{N(B)^{n_{GB}} + k_{GB}^{n_{GB}}} \cdot \frac{N(F)^{n_{GF}}}{N(F)^{n_{GF}} + k_{GF}^{n_{GF}}} \cdot \frac{N(A)^{n_{GA}}}{N(A)^{n_{GA}} + k_{GA}^{n_{GA}}} \right) \end{aligned} \right) - N(G)$$

Variável H

$$\frac{dH}{dt} = \frac{1}{\tau_H} \left(\frac{N(F)^{n_{HF}}}{N(F)^{n_{HF}} + k_{HF}^{n_{HF}}} - N(H) \right)$$

Variável I

$$\frac{dI}{dt} = \frac{1}{\tau_I} \left(\frac{N(G)^{n_{IG}}}{N(G)^{n_{IG}} + k_{IG}^{n_{IG}}} \cdot \frac{N(H)^{n_{IH}}}{N(H)^{n_{IH}} + k_{IH}^{n_{IH}}} - N(I) \right)$$

Variável J

$$\frac{dJ}{dt} = \frac{1}{\tau_J} \left(\frac{N(I)^{n_{JI}}}{N(I)^{n_{JI}} + k_{JI}^{n_{JI}}} - N(J) \right)$$

